



1. Sorting Lists with Custom Keys

Lambda functions can define custom sorting criteria for lists.

```
names = ["Alice", "Bob", "Charlie"]
sorted_names = sorted(names, key=lambda x: len(x))
print(sorted_names)
```

```
['Bob', 'Alice', 'Charlie']
```

2. Filtering Data with filter()

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Use lambda with filter() to select elements from an iterable based on a condition.

```
numbers = [1, 2, 3, 4, 5]
even_numbers = list(filter(lambda x: x % 2 == 0, numbers))
print(even_numbers)
```

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```
[2, 4]
```

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7 CREATIVE WAYS TO LEVERAGE PYTHON LAMBDA FUNCTIONS



3. Transforming Data with map()

Apply a transformation to each element in an iterable using lambda with map().

```
numbers = [1, 2, 3]
squared = list(map(lambda x: x**2, numbers))
print(squared)
```

```
[1, 4, 9]
```

4. Inline Function in Event Handlers

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Lambda functions are handy for short callback functions in GUI or event-driven programming.

```
from tkinter import Tk, Button
root = Tk()
button = Button(root, text="Click", command=lambda: print("Button clicked!"))
button.pack()
root.mainloop()
```

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```
Button clicked!
Button clicked!
Button clicked!
```

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5. Reducing Lists with reduce()

Use lambda with reduce() from functools to cumulatively apply an operation to an iterable.

```
from functools import reduce
numbers = [1, 2, 3, 4]
product = reduce(lambda x, y: x * y, numbers)
print(product)
```

24

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6. Creating Dynamic Function Arguments

Lambda functions can dynamically generate functions for use in higher-order functions or closures.

```
def multiplier(n):
    return lambda x: x * n
double = multiplier(2)
print(double(5))
```

10

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6. Creating Dynamic Function Arguments

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print(double(5))
```

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10

7. Conditional Logic in Short Computations

Lambda functions can handle simple conditional logic for quick calculations.

```
max_value = lambda x, y: x if x > y else y  
print(max_value(10, 20))
```

20

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